International Agricultural Technologies for Implementation in India

1. Drone-Based Farming

What It Is
Drones with special cameras fly over fields to check crop health and find problem spots that are hard to see. Some drones can also spray fertilizers or pesticides only where needed, saving time and money.

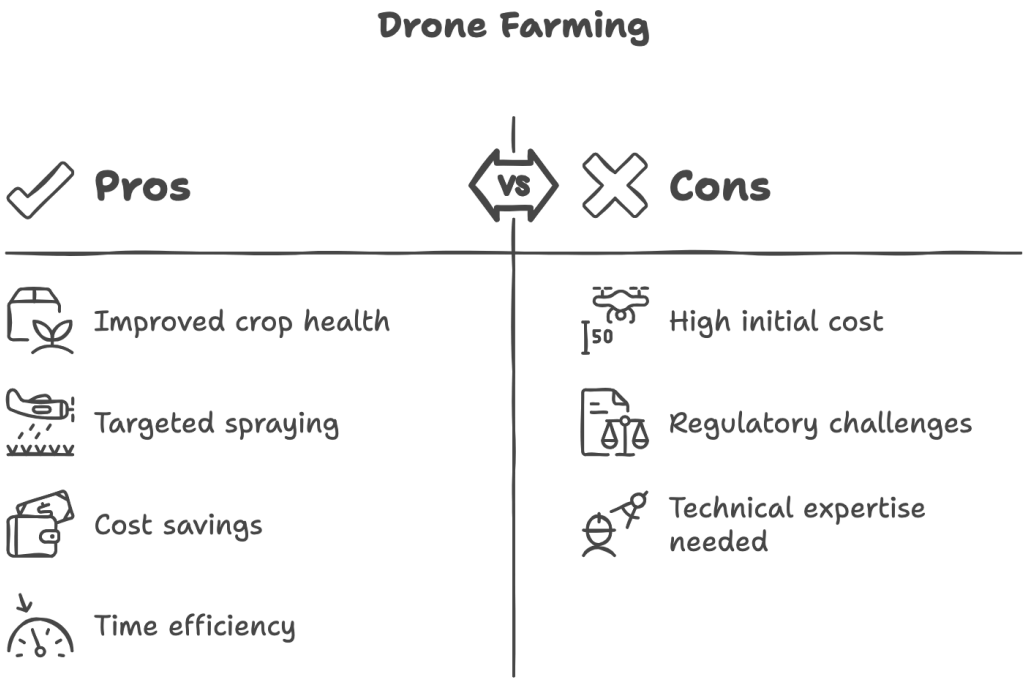
How It's Used Abroad

- In Brazil, soybean farmers use drones to cut pesticide use by about 30%, saving cash and helping the environment.
- Chinese rice farmers spot pests early with drones, protecting their crops.
- European vineyard owners use drones to know the best time to pick grapes, improving quality and profits. Do they use it to pick.??

Why India Needs It Now

- Farming costs are going up, and finding workers for planting or harvesting is tough. Drones make it easier to manage small fields quickly and accurately.
- The government offers subsidies, and drones are getting cheaper and easier to use, even for small farmers. [Consider avg income less than 100k.]

- With unpredictable weather, drones help catch issues like pests or dry soil early, so farmers can act fast.



2. Blockchain for Fair Markets

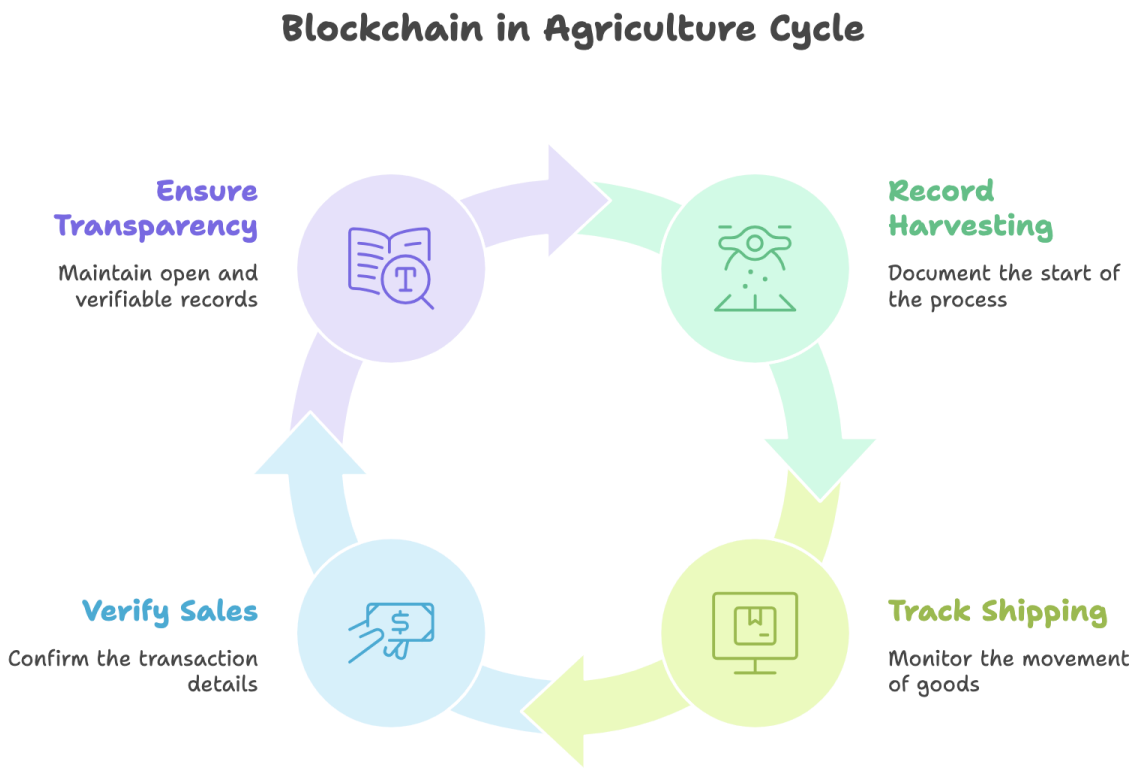
What It Is
Blockchain keeps a secure record of every step your produce takes, from the farm to the buyer. It tracks harvesting, shipping, and sales, so nothing can be hidden or faked.

How It's Used Abroad

- In China, blockchain cuts out middlemen, so farmers get paid in hours, not weeks.
- US farmers use it to prove their produce meets high standards, getting access to better markets.

Why India Needs It Now

- India's supply chains are messy, and farmers often get the smallest share of profits. Blockchain makes everything clear, reducing unfair pricing or cheating.
- Buyers, especially in cities or abroad, want to know where their food comes from. Blockchain proves it.
- With online markets growing in India, blockchain can help farmers gain trust and sell better globally.



3. Sensor-Based Vertical Farming (this is suitable for terrace farming)

What It Is

Crops grow in stacked layers inside controlled spaces, with sensors checking light, water, and nutrients all the time. These systems adjust conditions to keep plants healthy and growing well. It has potential in cities. than at villages.

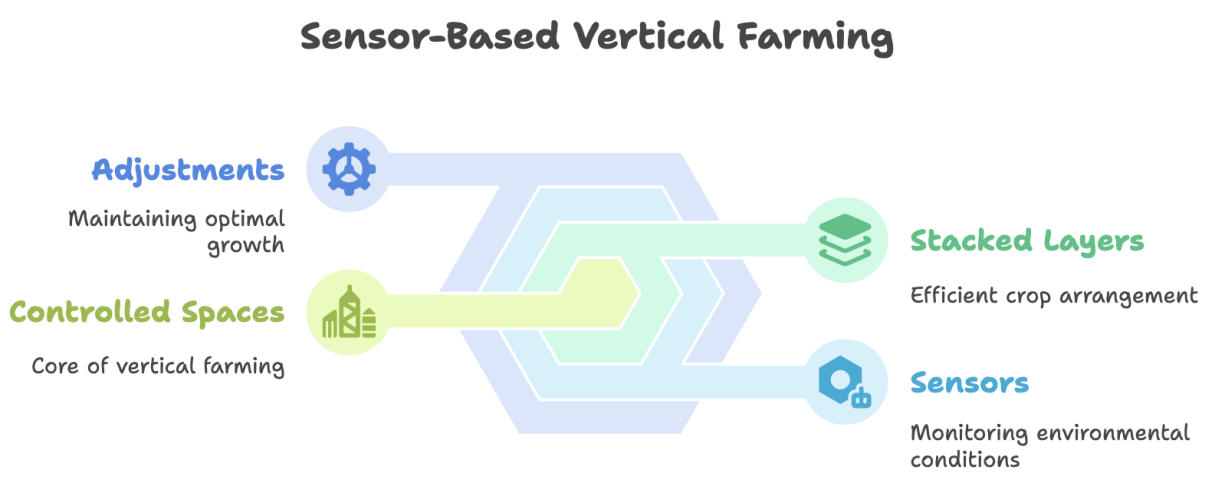
How It's Used Abroad

→ Effective at smaller land and limited to vegetables only.

- Singapore's vertical farms grow fresh greens for stores, using 95% less water than regular farming.
- The Netherlands turns old buildings into crop farms, growing valuable plants all year, no matter the weather.
- Japan uses sensors and AI in vertical farms to save labor and get high-quality crops every time.

Why India Needs It Now

- Cities are taking over farmland, but people in cities want fresh, chemical-free food.
- Unreliable rains and water shortages make regular farming risky. Vertical farms solve this and produce all year.
- Small vertical farms can be set up on rooftops or in empty buildings, creating jobs and bringing food closer to cities.
- With smart sensors, these farms are efficient and can start earning money quickly, even on a small scale.



4. Farmer Data Groups (this has huge potential)

What It Is

Farmers share info like soil tests, weather updates, and crop results on a digital platform. Smart tools turn this data into useful tips, like when to plant or how to handle pests.

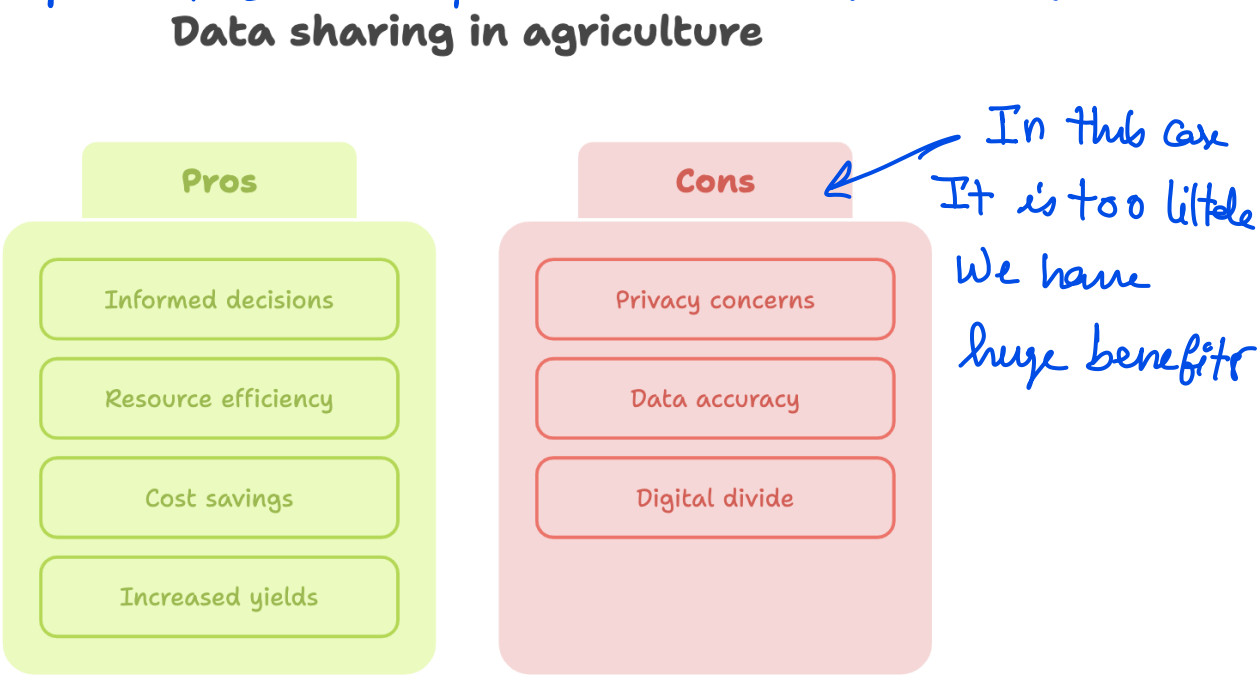
We are very scarce in these type of data.

How It's Used Abroad

* → Israel. ??

- In Denmark and the US, data groups help small farmers use the same tools as big companies, making things fairer.
- French farmers share data to get better deals on seeds and insurance, saving money.
- In the Netherlands, shared data has led to smarter farming, using less fertilizer without losing crops.

Why India Needs It Now

- Most Indian farmers work alone and don't have good, timely information, making planning hard.
- Joining data groups lets small farmers share tips, make better choices, and bargain together.
- India's Farmer Producer Organizations are ready to start these groups, connecting farmers who already trust each other.
- As more villages get internet, data-driven farming will be the future. Farmers who start now will stay ahead.

* Important to develop accurate forecast models